

OS Assignment 1

(C concepts revisit)

A) You are to write a C program that prints out all of its command-line arguments except those that begin with a dash. [Hint: find out about argc and argv.]

For example, after compiling your program into the file a.out, entering the command

a.out arg1 -arg2 arg3

prints

arg1 arg3

including a newline so your next shell prompt is not right after arg3. If all arguments begin with a dash, (or if there are no arguments at all) print nothing, that is, do not even print a newline. This is so that a blank line does not appear between the a.out command line and your next shell prompt.

(B) You are to write another C program which will read characters from its standard input stdin, count the number of NON-alphabetic ones including newlines (see the file /usr/include/ctype.h), that is, count those characters not in the a-z range nor in the A-Z range, and write out all characters read.

To read and write characters, this program uses only stdin and stdout, and only the stdio library routines (see stdio.h) for input and output (see getchar and putchar). When it hits EOF in its input, the program will print out the final non-alphabetic count on stderr using fprintf and then exit(0).

(C) You are to write yet another C program that reads characters from stdin, reverses lowercase and uppercase letters, that is, converts to lowercase all uppercase letters and vice versa (again, see ctype.h), and writes the results onto its standard output, stdout.

All characters read should be written whether converted or not. This program uses only stdin and stdout, and the stdio library routines. The program will exit(0) when it hits EOF in its input.